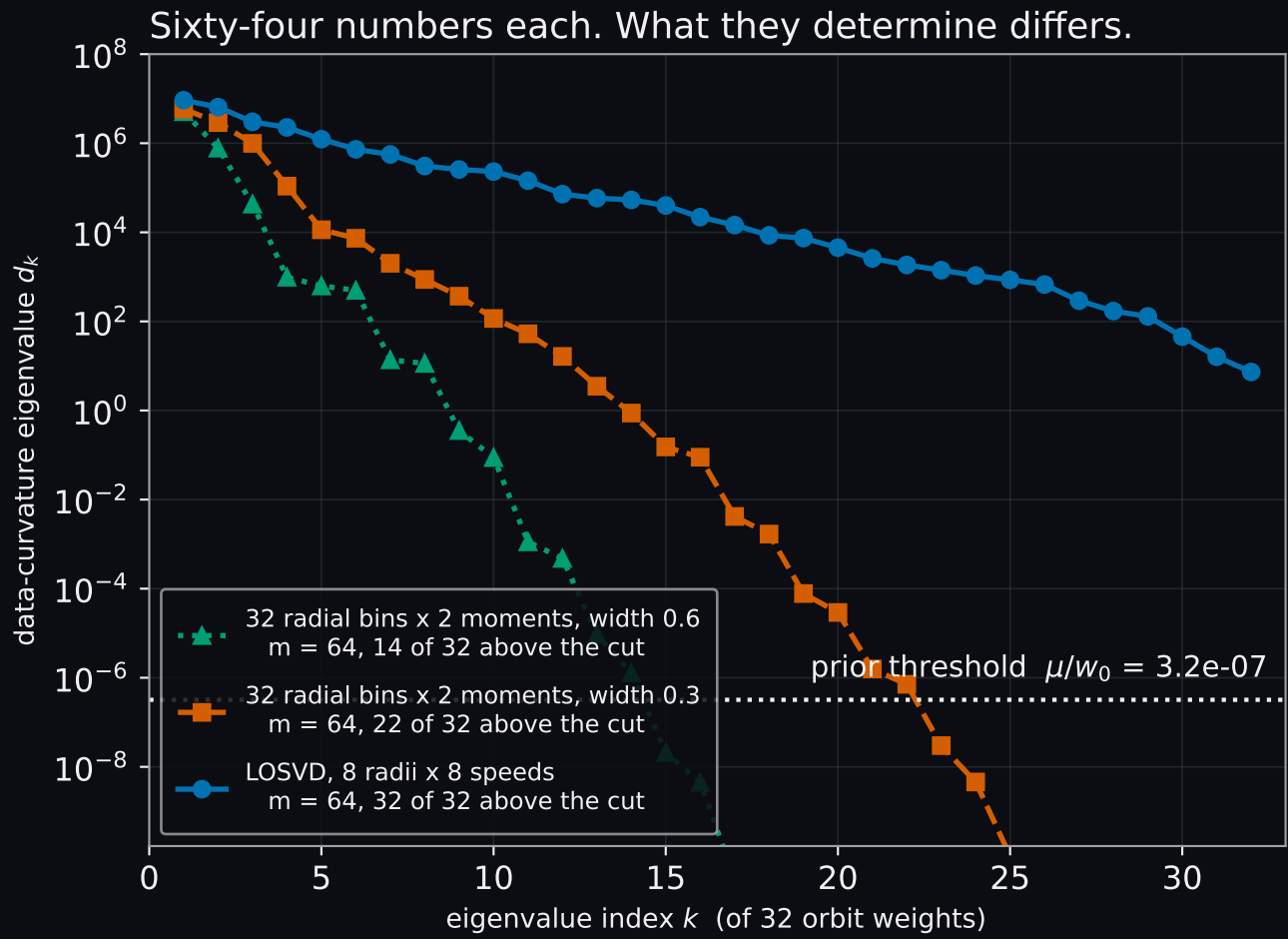




All three observable sets contain exactly 64 numbers, against 32 orbit weights. This matters: effective_parameters is bounded above by $\min(m, n)$, so comparing sets of different size would demonstrate only that bound. Holding m fixed, what the data determine ranges from 13.8 to 32.0 of the 32 weights according to what was measured. The radial sets place 32 Gaussian bins over 0.4 to 2.0 at a spacing of 0.05, so bins of width 0.3 and 0.6 overlap substantially. That redundancy is deliberate and is the variable under test. Note also that effective_parameters separates the under-determined from the determined case sharply, but does not finely predict the weight error within the under-determined regime, where the prior fills a null space of comparable size either way.

Mock tracer orbits in a softened point mass -- not a galaxy, not an instrument, not data. $N = 32$ orbits, softening = 0.2, 48 steps of $dt = 0.12$, entropy prior $\mu = 1e-08$, LBFGS for 1500 steps. Eigenvalues are the squared singular values of the whitened Jacobian, which is exact here and avoids the negative eigenvalues eigvalsh returns at this spectrum. effective_parameters is a local Gaussian-limit quantity, not a posterior.